

Elementary & Intermediate Algebra

ANSWER KEY



Solving linear equations

- 1 Solve for x : $4x - 9 = 23$.

$$4x = 32, \text{ so } x = 8.$$

- 2 Solve for x : $\frac{3x - 2}{5} = 8$.

Multiply both sides by 5: $3x - 2 = 40$, so $3x = 42$, giving $x = 14$.

Solving and graphing inequalities

- 3 Solve $5x + 3 < 28$, giving your answer as an inequality.

$$5x < 25, \text{ so } x < 5.$$

- 4 Solve $-4x + 7 \leq -9$, being careful with the direction of the inequality.

$$-4x \leq -16. \text{ Dividing by } -4 \text{ flips the inequality: } x \geq 4.$$

Systems of linear equations

- 5 Solve the system: $y = 3x - 4$ and $2x + y = 11$.

Substitute: $2x + (3x - 4) = 11$, so $5x - 4 = 11$, giving $5x = 15$, $x = 3$. Then $y = 3(3) - 4 = 5$. Solution: $(3, 5)$.

- 6 Solve the system: $3x + 4y = 18$ and $x - 2y = -4$, using elimination.

Multiply the second equation by 2: $2x - 4y = -8$. Add to the first: $5x = 10$, so $x = 2$. Substituting into $x - 2y = -4$: $2 - 2y = -4$, so $-2y = -6$, giving $y = 3$. Solution: $(2, 3)$.

Factoring and solving quadratic equations

- 7 Solve $x^2 - x - 12 = 0$ by factoring.

$$\text{Factor: } (x - 4)(x + 3) = 0, \text{ so } x = 4 \text{ or } x = -3.$$

- 8 Solve $2x^2 - 5x - 3 = 0$ by factoring.

$$\text{Factor: } (2x + 1)(x - 3) = 0, \text{ so } x = -\frac{1}{2} \text{ or } x = 3.$$

Simplifying rational and radical expressions

9 Simplify $\frac{x^2 - 9}{x + 3}$.

Factor the numerator: $\frac{(x - 3)(x + 3)}{x + 3} = x - 3$ (for $x \neq -3$).

10 Simplify $\sqrt{48}$.

$$\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3}.$$

Working with exponents

11 Simplify $\frac{x^7}{x^3}$.

$$\frac{x^7}{x^3} = x^{7-3} = x^4.$$

12 Simplify $(3x^2y)^3$.

$$(3x^2y)^3 = 3^3x^6y^3 = 27x^6y^3.$$

Absolute value equations

13 Solve $|3x + 2| = 11$.

Either $3x + 2 = 11$, giving $x = 3$, or $3x + 2 = -11$, giving $x = -\frac{13}{3}$.

Word problems leading to linear or quadratic equations

14 The length of a rectangle is 3 more than twice its width, and its area is 65. Find the width and length.

Let width = w , length = $2w + 3$. Area: $w(2w + 3) = 65$, so $2w^2 + 3w - 65 = 0$. Factor:
 $(2w + 13)(w - 5) = 0$, giving $w = 5$ (rejecting the negative solution $w = -6.5$). Length: $2(5) + 3 = 13$.

15 A number increased by 8 is 3 times the original number decreased by 4. Find the number.

Let the number be x : $x + 8 = 3x - 4$. Solving: $12 = 2x$, so $x = 6$.

Combining algebraic techniques in a multi-step problem

- 16** This question has two parts. Two consecutive positive integers have a product of 132. (a) Set up and solve a quadratic equation to find the two integers. (b) Verify your answer by checking the product.

(a) Let the integers be n and $n + 1$: $n(n + 1) = 132$, so $n^2 + n - 132 = 0$. Factor: $(n + 12)(n - 11) = 0$, giving $n = 11$ (rejecting the negative solution $n = -12$). The integers are 11 and 12. (b) Check: $11 \times 12 = 132$ ✓.

Solving quadratics using the quadratic formula

- 17** Solve $x^2 + 3x - 2 = 0$ using the quadratic formula, giving exact answers.

$$x = \frac{-3 \pm \sqrt{9 + 8}}{2} = \frac{-3 \pm \sqrt{17}}{2}.$$

- 18** Use the discriminant to determine the number of real solutions of $4x^2 - 12x + 9 = 0$, without solving.

Discriminant $= (-12)^2 - 4(4)(9) = 144 - 144 = 0$. Since the discriminant is zero, there is exactly one real (repeated) solution.

Factoring by grouping and special patterns

- 19** Factor $x^3 + 2x^2 - 9x - 18$ by grouping.

Group terms: $x^2(x + 2) - 9(x + 2) = (x + 2)(x^2 - 9)$. Factor the difference of squares: $(x + 2)(x - 3)(x + 3)$.

- 20** Factor $4x^2 - 25$ completely.

Difference of squares: $4x^2 - 25 = (2x - 5)(2x + 5)$.

Rational equations and extraneous solutions

- 21** Solve $\frac{x}{x-2} = \frac{3}{x-2} + 1$, checking for extraneous solutions.

Multiply through by $(x - 2)$: $x = 3 + (x - 2)$, so $x = x + 1$, giving $0 = 1$ — a contradiction, meaning there is no solution (this reflects that $x = 2$ would make the denominators undefined, and no other value works).

Multi-step algebraic word problems

- 22 This question has two parts. A boat travels 60 miles downstream in 3 hours and returns upstream in 5 hours. (a) Let b be the boat's speed in still water and c be the current's speed; write two equations based on the trip times. (b) Solve the system to find b and c .

(a) Downstream speed is $b + c$: $3(b + c) = 60$, so $b + c = 20$. Upstream speed is $b - c$: $5(b - c) = 60$, so $b - c = 12$. (b) Adding the two equations: $2b = 32$, so $b = 16$. Then $c = 20 - 16 = 4$. The boat's speed in still water is 16 mph, and the current's speed is 4 mph.

Multiple choice — easy

- 23 Solve for x : $3x + 4 = 31$.

- a. 10
- b. 8
- c. 9 ✓
- d. 11

$$3x = 27, \text{ so } x = 9.$$

- 24 Solve for x : $9x + 8 = -46$.

- a. -4
- b. -7
- c. -6 ✓
- d. -5

$$9x = -54, \text{ so } x = -6.$$

- 25 Solve for x : $8x + 14 = 86$.

- a. 9 ✓
- b. 11
- c. 10
- d. 8

$$8x = 72, \text{ so } x = 9.$$

26 Solve for x : $4x + 14 = -14$.

a. -5

b. -7 ✓

c. -6

d. -8

$4x = -28$, so $x = -7$.

27 Solve for x : $9x = 36$.

a. 4 ✓

b. 3

c. 5

d. 6

$9x = 36$, so $x = 4$.

28 Solve for x : $2x + 8 = 26$.

a. 8

b. 10

c. 11

d. 9 ✓

$2x = 18$, so $x = 9$.

29 Solve for x : $6x - 13 = 41$.

a. 8

b. 10

c. 9 ✓

d. 11

$6x = 54$, so $x = 9$.

30 Solve for x : $4x - 13 = 15$.

a. 8

b. 6

c. 7 ✓

d. 9

$4x = 28$, so $x = 7$.

31 Solve for x : $7x - 13 = 15$.

a. 6

b. 4 ✓

c. 5

d. 3

$7x = 28$, so $x = 4$.

32 Solve for x : $5x - 14 = 16$.

a. 7

b. 8

c. 6 ✓

d. 5

$5x = 30$, so $x = 6$.

33 Solve for x : $5x + 15 = -15$.

- a. -4
- b. -5
- c. -7
- d. -6 ✓

$$5x = -30, \text{ so } x = -6.$$

34 Solve for x : $9x + 17 = 125$.

- a. 13
- b. 12 ✓
- c. 11
- d. 14

$$9x = 108, \text{ so } x = 12.$$

35 Solve for x : $9x - 18 = 27$.

- a. 7
- b. 5 ✓
- c. 4
- d. 6

$$9x = 45, \text{ so } x = 5.$$

36 Solve for x : $6x = 48$.

a. 9

b. 7

c. 8 ✓

d. 10

$6x = 48$, so $x = 8$.

37 Solve for x : $5x + 17 = -33$.

a. -11

b. -10 ✓

c. -8

d. -9

$5x = -50$, so $x = -10$.

Multiple choice — medium

38 Solve $x^2 - 1 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. (1, -1)

b. (0, 1)

c. (-1, 1) ✓

d. (-1, 0)

Factor: $(x - (-1))(x - 1) = 0$, so $x = -1$ or $x = 1$. Answer: (-1, 1).

39 Solve $x^2 - 49 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(7, -7)$
- b. $(-7, 6)$
- c. $(-6, 7)$
- d. $(-7, 7)$ ✓

Factor: $(x - (-7))(x - 7) = 0$, so $x = -7$ or $x = 7$. Answer: $(-7, 7)$.

40 Solve $x^2 + 2x - 8 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(4, -2)$
- b. $(-3, 2)$
- c. $(-4, 1)$
- d. $(-4, 2)$ ✓

Factor: $(x - (-4))(x - 2) = 0$, so $x = -4$ or $x = 2$. Answer: $(-4, 2)$.

41 Solve $x^2 - 6x - 16 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(2, -8)$
- b. $(-1, 8)$
- c. $(-2, 7)$
- d. $(-2, 8)$ ✓

Factor: $(x - 8)(x - (-2)) = 0$, so $x = 8$ or $x = -2$. Answer: $(-2, 8)$.

42 Solve $x^2 - x - 42 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(6, -7)$
- b. $(-6, 7)$ ✓
- c. $(-5, 7)$
- d. $(-6, 6)$

Factor: $(x - (-6))(x - 7) = 0$, so $x = -6$ or $x = 7$. Answer: $(-6, 7)$.

43 Solve $x^2 + 4x - 60 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. $(-10, 6)$ ✓

b. $(-9, 6)$

c. $(-10, 5)$

d. $(10, -6)$

Factor: $(x - (-10))(x - 6) = 0$, so $x = -10$ or $x = 6$. Answer: $(-10, 6)$.

44 Solve $x^2 - 7x - 8 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. $(0, 8)$

b. $(-1, 7)$

c. $(1, -8)$

d. $(-1, 8)$ ✓

Factor: $(x - (-1))(x - 8) = 0$, so $x = -1$ or $x = 8$. Answer: $(-1, 8)$.

45 Solve $x^2 + 17x + 70 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. $(-10, -7)$ ✓

b. $(-9, -7)$

c. $(-10, -8)$

d. $(10, 7)$

Factor: $(x - (-7))(x - (-10)) = 0$, so $x = -7$ or $x = -10$. Answer: $(-10, -7)$.

46 Solve $x^2 + 4x - 21 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(-6, 3)$
- b. $(-7, 3)$ ✓
- c. $(7, -3)$
- d. $(-7, 2)$

Factor: $(x - (-7))(x - 3) = 0$, so $x = -7$ or $x = 3$. Answer: $(-7, 3)$.

47 Solve $x^2 - 4x - 5 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(1, -5)$
- b. $(-1, 5)$ ✓
- c. $(-1, 4)$
- d. $(0, 5)$

Factor: $(x - (-1))(x - 5) = 0$, so $x = -1$ or $x = 5$. Answer: $(-1, 5)$.

48 Solve $x^2 + x - 2 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(-1, 1)$
- b. $(-2, 0)$
- c. $(-2, 1)$ ✓
- d. $(2, -1)$

Factor: $(x - 1)(x - (-2)) = 0$, so $x = 1$ or $x = -2$. Answer: $(-2, 1)$.

49 Solve $x^2 - 11x + 18 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

- a. $(2, 9)$ ✓
- b. $(3, 9)$
- c. $(2, 8)$
- d. $(-2, -9)$

Factor: $(x - 9)(x - 2) = 0$, so $x = 9$ or $x = 2$. Answer: $(2, 9)$.

50 Solve $x^2 - 16x + 63 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. (7, 8)

b. (-7, -9)

c. (7, 9) ✓

d. (8, 9)

Factor: $(x - 7)(x - 9) = 0$, so $x = 7$ or $x = 9$. Answer: (7, 9).

51 Solve $x^2 - 3x = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. (1, 3)

b. (0, 2)

c. (0, -3)

d. (0, 3) ✓

Factor: $(x - 0)(x - 3) = 0$, so $x = 0$ or $x = 3$. Answer: (0, 3).

52 Solve $x^2 - 25 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. (-5, 4)

b. (5, -5)

c. (-4, 5)

d. (-5, 5) ✓

Factor: $(x - (-5))(x - 5) = 0$, so $x = -5$ or $x = 5$. Answer: (-5, 5).

Multiple choice — hard

53 Solve $x^2 + 8x - 8 = 0$ using the quadratic formula, giving exact answers.

a. $-4 \pm 2\sqrt{6}$ ✓

b. $-3 \pm 2\sqrt{6}$

c. $-4 \pm 3\sqrt{6}$

d. $4 \pm 2\sqrt{6}$

$$x = \frac{-8 \pm \sqrt{(8)^2 - 4(-8)}}{2} = \frac{-8 \pm \sqrt{96}}{2} = -4 \pm 2\sqrt{6}.$$

54 Solve $x^2 + 14x - 1 = 0$ using the quadratic formula, giving exact answers.

a. $-6 \pm 5\sqrt{2}$

b. $-7 \pm 5\sqrt{2}$ ✓

c. $7 \pm 5\sqrt{2}$

d. $-7 \pm 6\sqrt{2}$

$$x = \frac{-14 \pm \sqrt{(14)^2 - 4(-1)}}{2} = \frac{-14 \pm \sqrt{200}}{2} = -7 \pm 5\sqrt{2}.$$

55 Solve $x^2 - 10x - 9 = 0$ using the quadratic formula, giving exact answers.

a. $6 \pm \sqrt{34}$

b. $5 \pm 2\sqrt{34}$

c. $5 \pm \sqrt{34}$ ✓

d. $-5 \pm \sqrt{34}$

$$x = \frac{10 \pm \sqrt{(-10)^2 - 4(-9)}}{2} = \frac{10 \pm \sqrt{136}}{2} = 5 \pm \sqrt{34}.$$

56 Solve $x^2 - 10x + 4 = 0$ using the quadratic formula, giving exact answers.

a. $6 \pm \sqrt{21}$

b. $5 \pm \sqrt{21}$ ✓

c. $5 \pm 2\sqrt{21}$

d. $-5 \pm \sqrt{21}$

$$x = \frac{10 \pm \sqrt{(-10)^2 - 4(4)}}{2} = \frac{10 \pm \sqrt{84}}{2} = 5 \pm \sqrt{21}.$$

57 Solve $x^2 + 2x - 2 = 0$ using the quadratic formula, giving exact answers.

a. $1 \pm \sqrt{3}$

b. $-1 \pm 2\sqrt{3}$

c. $-1 \pm \sqrt{3}$ ✓

d. $0 \pm \sqrt{3}$

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(-2)}}{2} = \frac{-2 \pm \sqrt{12}}{2} = -1 \pm \sqrt{3}.$$

58 Simplify $(x^2)^3$.

a. x^8

b. x^6 ✓

c. x^5

d. x^7

$$(x^2)^3 = x^{2 \times 3} = x^6.$$

59 Simplify x^7/x^3 .

a. x^5

b. x^3

c. x^4 ✓

d. x^6

$$x^7/x^3 = x^{7-3} = x^4.$$

60 Simplify x^{10}/x^9 .

a. x^2

b. x^3

c. x^1 ✓

d. x^4

$$x^{10}/x^9 = x^{10-9} = x^1.$$

61 Simplify x^9/x^7 .

a. x^3

b. x^4

c. x^2 ✓

d. x^1

$$x^9/x^7 = x^{9-7} = x^2.$$

62 Simplify $(x^4)^3$.

a. x^{12} ✓

b. x^{14}

c. x^{11}

d. x^{13}

$$(x^4)^3 = x^{4 \times 3} = x^{12}.$$

63 Solve the system: $3x + 4y = 26$ and $5x + y = 15$. Find the value of x .

a. 1

b. 2 ✓

c. 5

d. 3

Solving the system gives $x = 2$, $y = 5$. Check: $3(2) + 4(5) = 26$ ✓, $5(2) + (5) = 15$ ✓.

64 Solve the system: $x + 6y = -41$ and $2x + y = -16$. Find the value of x .

- a. -3
- b. -4
- c. -6
- d. -5 ✓

Solving the system gives $x = -5$, $y = -6$. Check: $(-5) + 6(-6) = -41$ ✓, $2(-5) + (-6) = -16$ ✓.

65 Solve the system: $2x + 3y = 2$ and $4x + y = 14$. Find the value of x .

- a. -2
- b. 5
- c. 4 ✓
- d. 3

Solving the system gives $x = 4$, $y = -2$. Check: $2(4) + 3(-2) = 2$ ✓, $4(4) + (-2) = 14$ ✓.

66 Solve the system: $3x + 6y = 3$ and $6x + 3y = -12$. Find the value of x .

- a. -2
- b. -3 ✓
- c. -4
- d. 2

Solving the system gives $x = -3$, $y = 2$. Check: $3(-3) + 6(2) = 3$ ✓, $6(-3) + 3(2) = -12$ ✓.

67 Solve the system: $2x + 4y = 18$ and $6x + 4y = 22$. Find the value of x .

a. 2

b. 4

c. 0

d. 1 ✓

Solving the system gives $x = 1$, $y = 4$. Check: $2(1) + 4(4) = 18$ ✓, $6(1) + 4(4) = 22$ ✓.